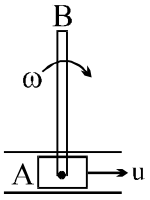
SECTION-I

Single Correct Answer Type **5 Q. [3 M (–1)]**

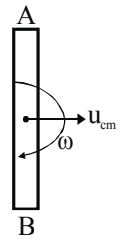
1. A mechanism consists of a part which is translated with a velocity u and a rod AB of length L and mass M hinged at A. The rod rotates about axis A with angular velocity ω. The kinetic energy of rod when it is vertical as shown is



(A) $\frac{1}{2}Mu^2 + \frac{1}{6}ML^2\omega^2$ (B) $\frac{1}{2}Mu^2 + \frac{1}{6}ML\omega u$

(C) $\frac{1}{2}Mu^2 + \frac{1}{6}ML^2\omega^2 + \frac{1}{2}ML\omega u$ (D) None of these

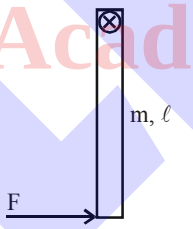
2. A rod of length ℓ is rotating with angular velocity ω and translating such that $u_{cm} = \omega\ell/2$. The distance covered by the point B when the rod completes one full rotation is



(A) $\pi\ell$ (B) 8ℓ

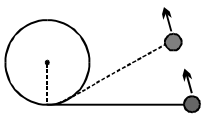
(C) 4ℓ (D) $2\pi\ell$

3. As shown in the figure rod of mass m and length ℓ is hinged at O at one end and free to rotate in vertical plane. A force F of magnitude mg is applied on another end of rod which always act perpendicular to the rod. Force is start acting at time t = 0 (as shown). Find vertical component of normal reaction acting on rod when rod become horizontal for 3rd time:- (given m = 2 kg, ℓ = 1m, g = 10 m/s²)



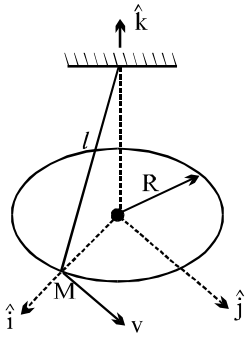
(A) 5 N (B) 10N (C) 15 N (D) 20 N

4. A ball is attached to a string that is attached to a pole. When the ball is hit, the string wraps around the pole and the ball spiral inwards sliding on the frictionless surface. Neglecting air resistance, what happens as the ball swings around the pole? Consider angular momentum about centre of pole.


Top View

(A) The mechanical energy and angular momentum are conserved.
(B) The angular momentum of the ball is conserved but the mechanical energy of the ball increases.
(C) The angular momentum of the ball is conserved but the mechanical energy of the ball decreases.
(D) The mechanical energy of the ball is conserved but angular momentum of the ball decreases.

5. A conical pendulum consists of a mass M suspended from a string of length ℓ. The mass executes a circle of radius R in a horizontal plane with speed v. At time t, the mass is at position $R\hat{i}$ and has velocity $v\hat{j}$. At time t, the angular momentum vector of the mass M about the point from which the string suspended is :



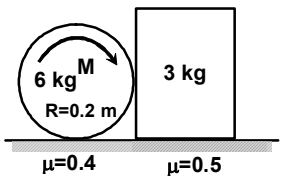
(A) $MvR\hat{k}$ (B) $Mv\ell\hat{k}$

(C) $Mv\ell\left[\frac{\sqrt{\ell^2 - R^2}}{\ell}\hat{i} + \frac{R}{\ell}\hat{k}\right]$ (D) $-Mv\ell\left[\frac{\sqrt{\ell^2 - R^2}}{\ell}\hat{i} + \frac{R}{\ell}\hat{k}\right]$

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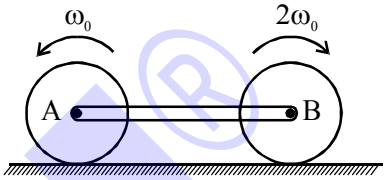
Multiple Correct Answer Type 8 Q. [4 M (–1)]

6. A clockwise torque of 6N-m is applied to the circular cylinder as shown in the figure. There is no friction between the cylinder and the block.



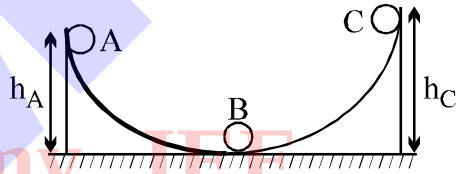
(A) The cylinder will be slipping but the system does not move forward
(B) The system cannot move forward for any torque applied to the cylinder
(C) The acceleration of the system will be $1 m/s^2$ forward
(D) The angular acceleration of the cylinder is $10 rad/s^2$.

7. Two identical disc having radius R are rotating in opposite sense are joined with light rod having frictionless groove A and B and placed on rough horizontal surface as shown in figure. Initially translation velocity of each disc is zero. Then :-



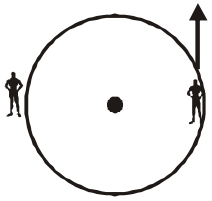
(A) Final velocity of B is $\frac{R\omega_0}{6}$
(B) Rod's length will increase initially due to stress
(C) Finally rod has tensile stress
(D) Mechanical energy will decrease

8. A ball moves over a fixed track as shown in the figure. From A to B the ball rolls without slipping. Surface BC is frictionless. K_A, K_B and K_C are kinetic energies of the ball at A, B and C, respectively. Then



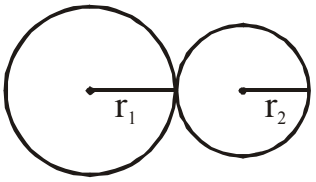
(A) $h_A > h_C$; $K_B > K_C$
(B) $h_A > h_C$; $K_C > K_A$
(C) $h_A = h_C$; $K_B = K_C$
(D) $h_A < h_C$; $K_B > K_C$

9. A student Amir of mass m is standing on the edge of an horizontal disk of radius R. The disc is free to rotate about a frictionless vertical axis passing through its centre. Initially the student and disc are at rest. At t = 0 the student starts to run at his maximum velocity v relative to disc. He runs along the circumference of the disc towards his friend Salman in anticlockwise sense. Salman is stationary on ground. Moment of inertia of disc is I.



(A) Time taken to reach Salman is $\frac{\pi R}{v}\left(1+\frac{I}{mR^2}\right)$.
(B) If disc were considered massless time taken to reach Salman is $\frac{\pi R}{v}$.
(C) If disc were considered massless Amir can never reach Salman.
(D) Time taken will be minimum if Amir instead of running along circumference runs along diameter towards Salman.

10. Two horizontal discs of different radii are free to rotate about their central vertical axes. One is given some angular velocity, the other is stationary.Their rims are now brought in contact. There is friction between the rims :-



(A) The force of friction between the rims will disappear when the discs rotate with equal angular speeds
(B) The force of friction between the rims will disappear when they have equal linear velocities at the point of contact
(C) The angular momentum of the system will be conserved.
(D) The rotational kinetic energy of the system will not be conserved.

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11. A uniform solid cylinder of mass m and radius R is released from rest on a sufficiently rough inclined plane. During its downward journey along the incline, the cylinder moves distance ℓ along the incline. The angle of inclination from horizontal is α. Mark **CORRECT** statements :-

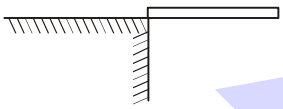
(A) The acceleration of centre of mass of cylinder is $\frac{2g\sin\alpha}{3}$.

(B) The final angular speed of cylinder is $\sqrt{\frac{4}{3}\frac{g\ell\sin\alpha}{R^2}}$.

(C) The minimum coefficient of friction required so that there is no slipping is $\frac{\tan\alpha}{3}$.

(D) If released from rest cylinder will undergo pure rolling even on smooth inclined plane.

12. A uniform rod of length l is falling down with a velocity v_0 when one of its end hits a fixed edge as shown.



(A) If the collision is elastic, the centre of mass may have upwards velocity just after the collision
(B) If the collision is partially elastic, the centre of mass may come to rest just after the collision
(C) If the collision is perfectly inelastic, the centre of mass has a downward velocity just after the collision
(D) If the collision is elastic, the centre of mass has a downward velocity just after the collision

13. Suppose a solid disk having radius R is put on a rough horizontal surface and given an angular speed ω_0 about an axis through it's centre and released. If coefficient of friction between disc and surface is μ then:

(A) Time interval before rolling motion starts is $\frac{R\omega_0}{3\mu g}$.

(B) Distance travelled by disc before pure rolling starts is $\frac{R^2\omega_0^2}{18\mu g}$.

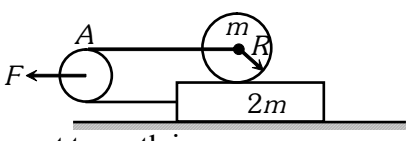
(C) Rolling motion doesn't occur

(D) Time interval before rolling motion starts is $\frac{2R\omega_0}{3\mu g}$.

Linked Comprehension Type (4 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question Nos. 14 to 16

A solid cylinder of mass m and radius R is kept at rest on a plank of mass 2m lying on a smooth horizontal surface. Massless and inextensible string connecting cylinder to the plank is passing over a massless pulley. The friction between the cylinder and the plank is sufficient to prevent slipping. Pulley A is pulled with a constant horizontal force F.



14. Acceleration of cylinder with respect to earth is

(A) $\frac{5F}{21m}$ (B) $\frac{F}{7m}$ (C) $\frac{3F}{7m}$ (D) $\frac{2F}{7m}$

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15. Acceleration of plank with respect to earth is

(A) $\frac{5F}{21m}$ (B) $\frac{F}{7m}$ (C) $\frac{3F}{7m}$ (D) $\frac{2F}{7m}$

16. Magnitude of friction force acting on the plank is

(A) $\frac{F}{7}$ (B) $\frac{F}{14}$ (C) $\frac{F}{21}$ (D) $\frac{2F}{7}$

Paragraph for Question Nos. 17 to 19

Greek lores have a fable of Sisyphus, a celebrated philosopher. He was punished in Hades for rebelling against the king. The punishment was to roll a spherical stone up a hill from where it would invariably roll down to the other side. He would run after the stone, to catch it back and roll it back uphill. The top of the hill was always a position of unstable equilibrium. The sphere and Sisyphus have same mass. Being a great philosopher, Sisyphus compared this act with our attempts to succeed in life. No matter how high we go, the stone of success always rolls down back.
Philosophy apart, answer the following questions.



17. At what height from the centre should sisyphus push in a direction parallel to hill to roll the sphere slowly with minimum force.

(A) 0 (B) R/2 (C) 2/5 R (D) R

18. If the sphere is rolling down, and Sisyphus wants to run along with the same acceleration without exerting any force on it.

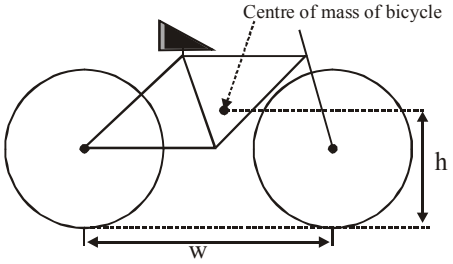
(A) friction on Sisyphus would be backwards and equal to that on sphere
(B) friction on Sisyphus would be forwards and greater than on sphere
(C) friction on Sisyphus would be backwards and less than on sphere
(D) friction on Sisyphus would be forwards and equal to that on sphere

19. What is the minimum friction coefficient on the hill for Sisyphus to just push the sphere up (assume it to be same for Sisyphus and Rock.) He pushes in a direction parallel to the hill.

(A) $\frac{3}{2}\tan\theta$ (B) $\tan\frac{\theta}{2}$ (C) $\frac{2}{7}\tan\theta$ (D) $\frac{5}{7}\tan\theta$

Paragraph for Questions no. 20 to 22

A simplified model of a bicycle of mass M has two tyres that each comes into contact with the ground at a point. The wheelbase of this bicycle (the distance between the points of contact with the ground) is w, and the center of mass of the bicycle is located midway between the tyres and at height h above the ground. Air resistance may be ignored.



Case -1 (Q. 20): Assume that cycle is moving rightwards & has acceleration a. Both tyres roll purely.

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20. The direction of friction on the tyres is

(A) forward on both (B) backward on both

(C) forward on front & backward on rear (D) forward on rear & backward on front.

Case-2 (Q. 21 and 22) : The bicycle is moving to the right, but slowing down at a constant rate. The acceleration has a magnitude a. Assume that the coefficient of sliding friction between each tyre and the ground is μ, and that both tyres are skidding (sliding without rotating). Express your answers in terms of w, h, M and g.

21. What is the maximum value of μ so that both tyres remain in contact with the ground?

(A) $\frac{w}{2h}$ (B) $\frac{h}{2w}$ (C) $\frac{2h}{w}$ (D) $\frac{w}{h}$

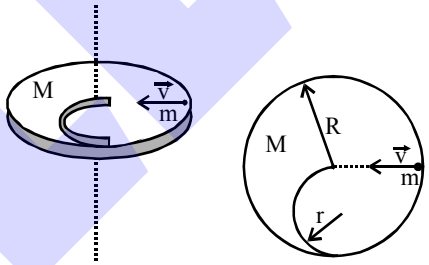
22. What is the maximum value of a so that both tyres remain in contact with the ground?

(A) $\frac{wg}{h}$ (B) $\frac{wg}{2h}$ (C) $\frac{hg}{2w}$ (D) $\frac{2hg}{w}$

Paragraph for Question Nos. 23 to 25

A smooth disc of mass M and radius R can rotate freely around a vertical axis supported by frictionless bearings. A constraining vertical surface of negligible mass, whose shape is a semicircular arc of radius $r = \frac{R}{2}$, is fixed on the disc as shown in the figure. A small ball of mass m is placed on the stationary disc and is bowled at a speed v in such a way that it reaches the internal side of the constraining surface tangentially.

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23. Mark the **CORRECT** statement :-

(A) Momentum of ball & disc system is conserved

(B) Mechanical energy of ball & disc system is not conserved

(C) Angular momentum of ball & disc system is conserved

(D) Mechanical energy of ball & disc system is not conserved due to impulsive reaction from support bearings

24. What is speed of the ball when it leaves the disc :-

(A) $v\sqrt{\frac{M}{M+2m}}$ (B) $v\sqrt{\frac{m}{M+2m}}$ (C) $v\sqrt{\frac{2M}{M+m}}$ (D) $2v\sqrt{\frac{M}{2M+m}}$

25. What is angular velocity of disc when ball leaves it :-

(A) $2\frac{mv}{MR}\sqrt{\frac{M}{M+2m}}$ (B) $2\frac{mv}{MR}\sqrt{\frac{m}{M+2m}}$ (C) $2\frac{mv}{MR}\sqrt{\frac{2M}{M+m}}$ (D) $2\frac{mv}{MR}\sqrt{\frac{M}{2M+m}}$

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SECTION-II

Numerical Answer Type Question **1Q.[3M (0)]**

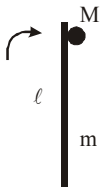
(upto second decimal place)

26. A solid cylinder rolls without slipping along a horizontal plane with velocity 66 m/s and reaches a plane inclined at 60° with the horizontal. If there is sufficient friction to prevent slipping then the body starts rolling up the inclined plane with the velocity v. Find the value of v (in m/s).

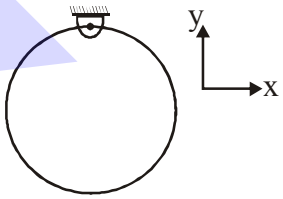
SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **3 Q. [4 M (0)]**

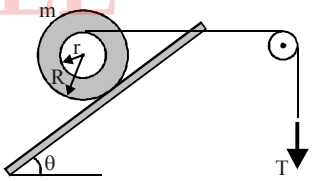
27. A uniform stick of mass 18 kg and length 3 m spins around on a frictionless horizontal plane, with its Centre of Mass stationary. A mass M is placed on the plane, and the stick collides elastically with it, as shown (with the contact point being the end of the stick). What should M (in kg) be so that after the collision the stick has translational motion, but no rotational motion?



28. A bullet of mass m moving with velocity $v_0(-\hat{k})$ strikes the bottom of a stationary vertical uniform ring of same mass m and radius R = 1 m. The ring lies in XY plane with its topmost point hinged on the ceiling. The ring can rotate about X-axis. There is no friction between the hinge and the ring. The bullet gets embedded in the ring immidiately after collision. Find the angular velocity of the system (in radian/s) just after collision. [Take v_0 = 11 m/s]

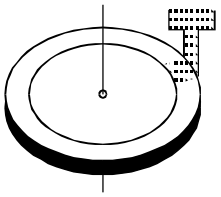


29. A spool, initially at rest, is kept on a frictionless incline making an angle θ = 37° with the horizontal. The mass of the spool is 2kg and it is pulled by a string as shown with a force of T = 10N. The string connecting the spool and the pulley is initially horizontal. Find the initial acceleration of the spool on the incline. Express your answer in m/s^2.



Subjective Type **1 Q. [4 M (0)]**

30. A uniform circular stone of mass M and radius R rotates around a vertical axle through its centre at an angular velocity ω_0 at time t = 0 as shown in the figure. The stone has a deep groove along its circumference. Sand is dumped into this groove at a constant rate of q. Sand does not spill out or slip relative to the stone once it falls into the groove.



(i) Assuming there is no external torque, find the angular velocity of the stone as a function of time t.
(ii) How much external torque would be needed to maintain the stone at a constant angular velocity ?

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GR # ROTATIONAL DYNAMICS-2 **ANSWER KEY**

SECTION-I

Single Correct Answer Type **5 Q. [3 M (–1)]**

1. Ans. (C) **2. Ans. (C)** **3. Ans. (C)** **4. Ans. (D)** **5. Ans. (C)**

Multiple Correct Answer Type **8 Q. [4 M (–1)]**

6. Ans. (C, D) **7. Ans. (A, B, D)** **8. Ans. (A,B)** **9. Ans. (C,D)** **10. Ans. (B, D)**

11. Ans. (A, B, C) **12. Ans. (C,D)** **13. Ans. (A, B)**

Linked Comprehension Type **(4 Para × 3Q.)** **[3 M (-1)]**

(Single Correct Answer Type)

14. Ans. (C) **15. Ans. (D)** **16. Ans. (B)** **17. Ans. (D)** **18. Ans. (A)** **19. Ans. (A)**

20. Ans. (D) **21. Ans. (A)** **22. Ans. (B)** **23. Ans. (C)** **24. Ans. (A)** **25. Ans. (A)**

SECTION-II

Numerical Answer Type Question **1Q.[3M(0)]**

(upto second decimal place)

26. Ans. 44

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **3 Q. [4 M (0)]**

27. Ans. 9 **28. Ans. 4** **29. Ans. 2**

Subjective Type **1 Q. [4 M (0)]**

30. Ans. (i) $\frac{M\omega_0}{M+2qt}$, (ii) $q\omega_0 R^2$

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